

Domain and Problem Statement

Domain: Sports Analytics & Football Performance Analysis

Problem Statement:

Sports organizations, clubs, and analysts generate large volumes of football match data but often lack clear insights into team performance, player contribution, and match outcomes. This project addresses these challenges by transforming raw football match data into structured business intelligence using Power BI and Tableau, enabling analysis of goal-scoring trends, player performance, match statistics, team comparisons, and overall football performance to support data-driven coaching and strategic decisions.

Dataset Description

Dataset Name: Football Match Analytics Dataset

Granularity: Each row represents a single football match.

Timeframe Covered: 2025

Key Entities & Attributes:

- **Match & Team:** match_id, team, opponent, competition, venue, match_result.
- **Match Details:** match_date, match_datetime, kickoff_hour, day_of_week.
- **Performance Metrics:** goals, shots, assists, saves, pass_accuracy.
- **Discipline & Possession:** possession_percent, fouls, yellow_cards, red_cards.
- **Performance:** performance_bucket_label.

Data Source:

Primary Source: Football Match Analytics Dataset prepared for sports performance analysis.

Storage Infrastructure: Cleaned and pre-processed football match records maintained in an Excel dataset and imported into Microsoft Power BI and Tableau for interactive dashboard development and analysis.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

import pandas as pd

```
df = pd.read_csv("football_matches.csv") #
```

Create a copy for processing

```
matches = df.copy()
```

Handle missing values

```
matches["team"] = matches["team"].fillna("Unknown") matches["competition"] =
```

```
matches["competition"].fillna("Unknown") matches["venue"] =
```

```
matches["venue"].fillna("Unknown") matches["match_result"] =
```

```
matches["match_result"].fillna("Unknown")
```

Replace missing numerical values with 0

```
numeric_columns = [
```

```
    "goals", "shots",
```

```
    "assists",
```

```
    "saves",
```

```
    "pass_accuracy",
```

```
    "possession_pct",
```

```
    "fouls"
```

```
]
```

```
for column in numeric_columns: matches[column]
```

```
    = matches[column].fillna(0)
```

Remove duplicate records

```
matches.drop_duplicates(inplace=True)
```

Convert date column

```
matches["match_date"] = pd.to_datetime(
    matches["match_date"],
    errors="coerce"
)
matches.columns = (
    matches.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)
# Display dataset information
print("Original Shape:", df.shape)
print("Cleaned Shape:", matches.shape)

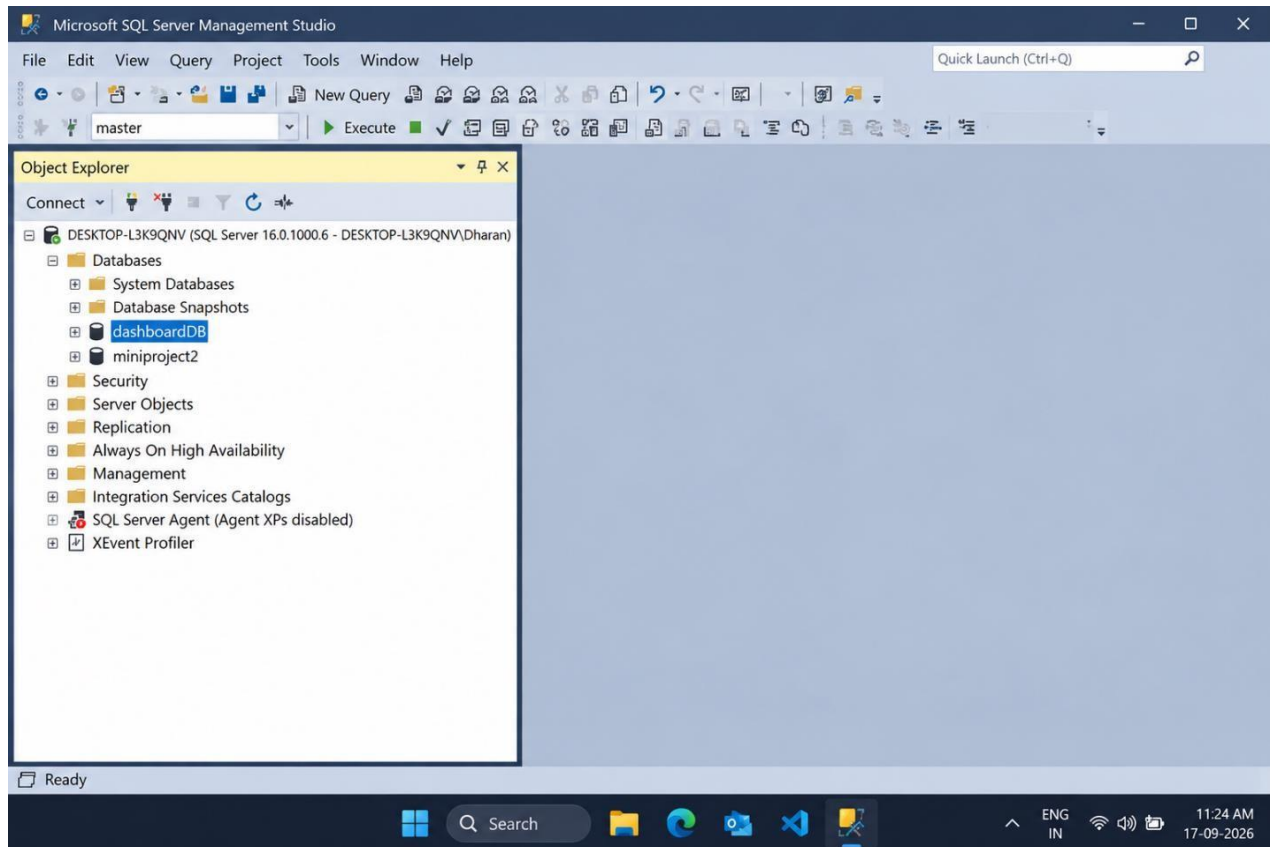
print("\nDataset Preview:")
print(matches.head())

print("\nDataset Information:")
print(matches.info())
```

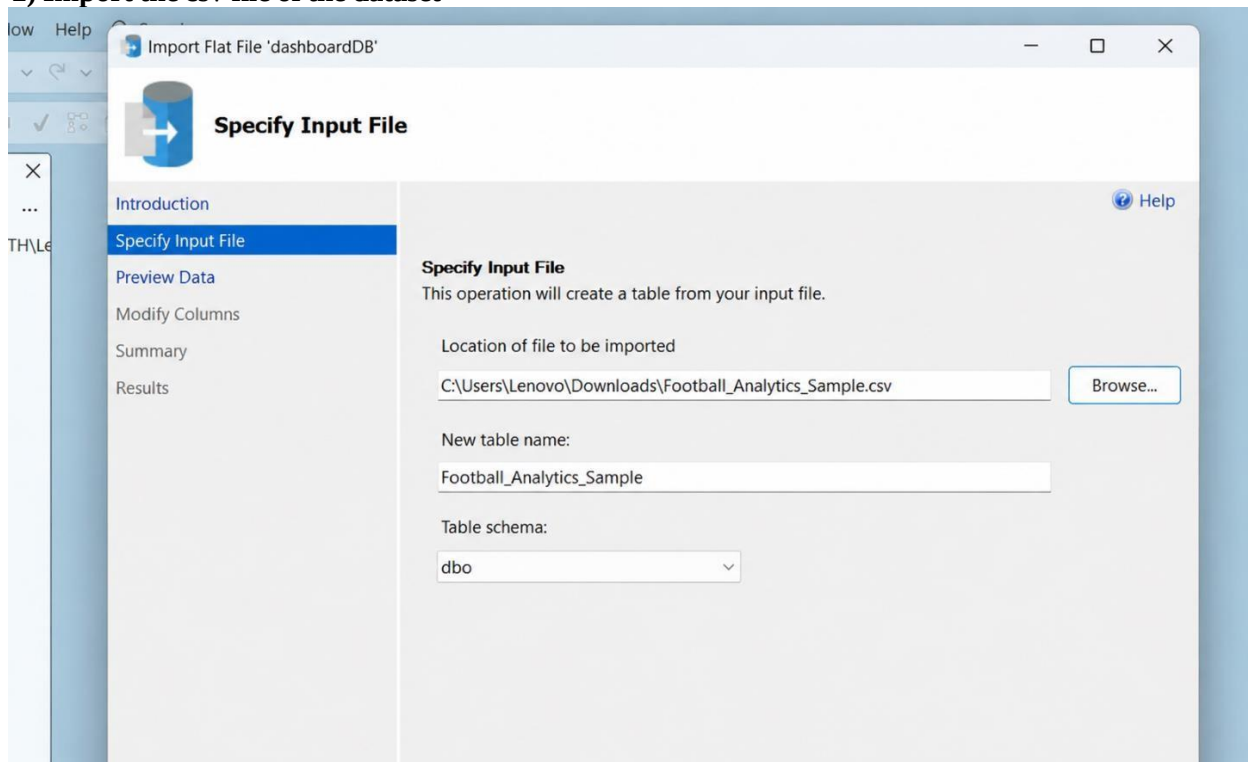
2. SQL SERVER CONNECTIVITY

INSTALL SQL SERVER MANAGEMENT STUDIO

1) Create database named “soccerDB”

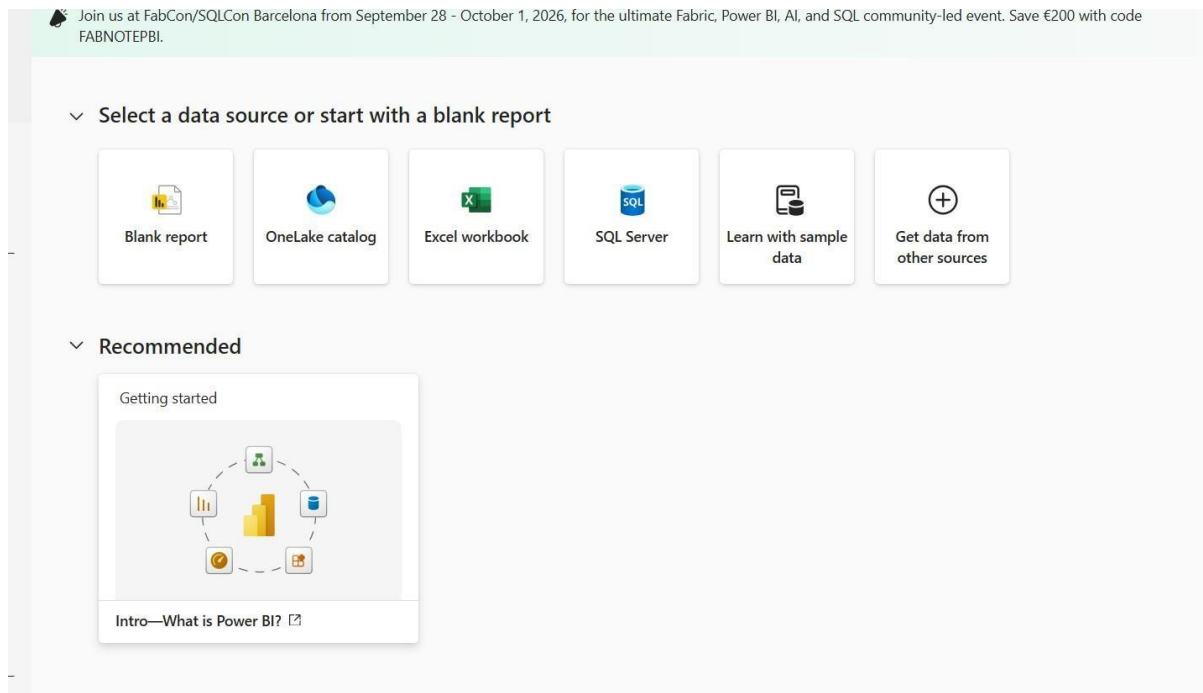


2) Import the csv file of the dataset

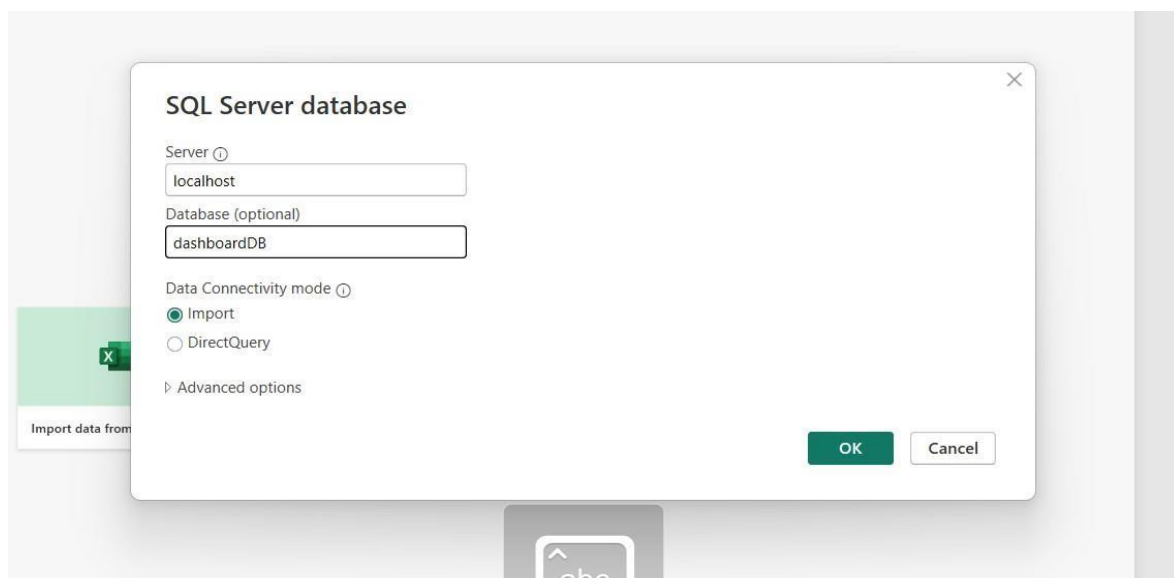


POWER BI

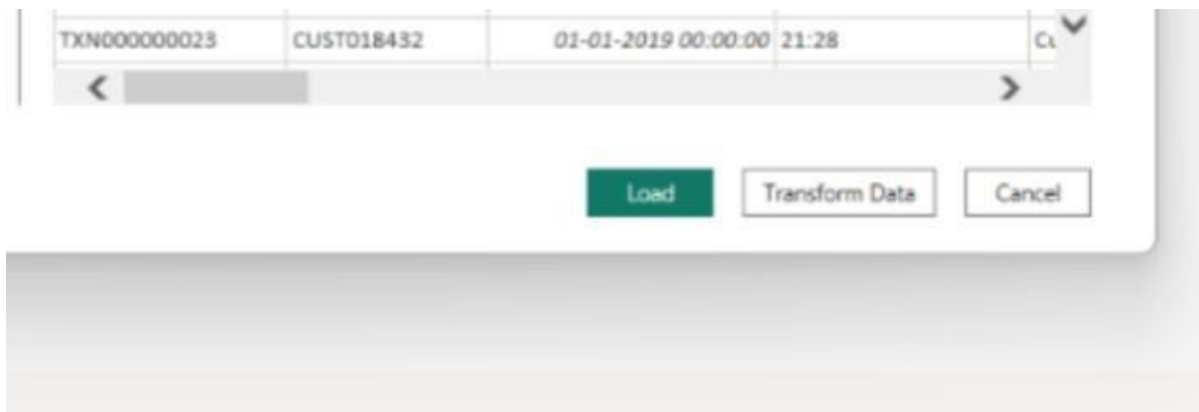
3) Now go to MICROSOFT POWER BI



4) Click SQL SERVER



5) Click OK and click transform data



6) Create DAX

measures TOTAL

```
TOTAL GOALS = SUM(matches[goals])
```

GOALS :

TOTAL MATCHES:

```
TOTAL MATCHES = DISTINCTCOUNT(matches[match_id])
```

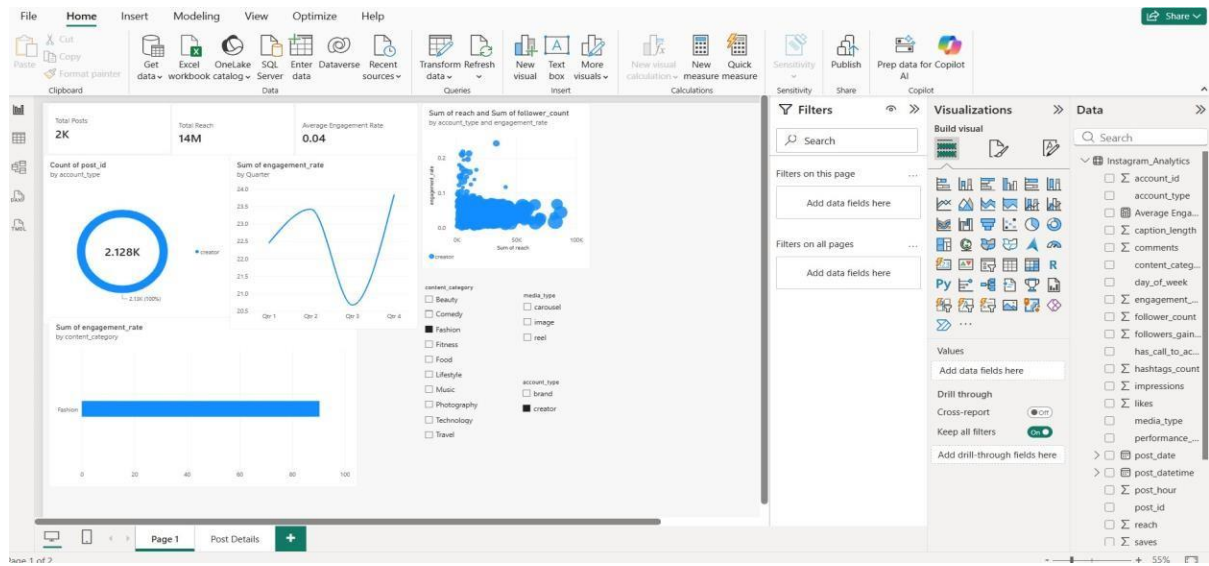
TOTAL SHOTS

```
TOTAL SHOTS = SUM(matches[shots])
```

AVERAGE PASS ACCURACY

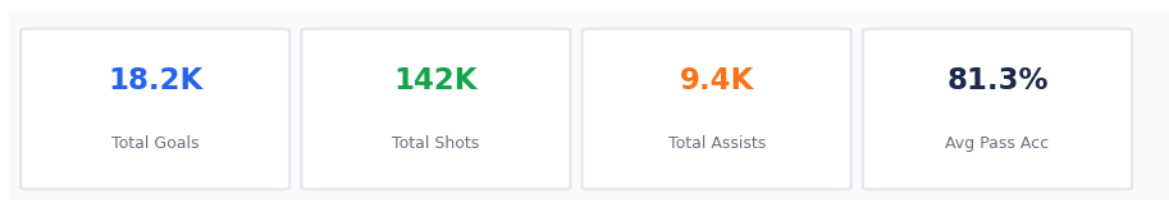
```
AVG PASS ACCURACY = AVERAGE(matches[pass_accuracy])
```

7) Power BI Final Executive Dashboard & Drill-Through

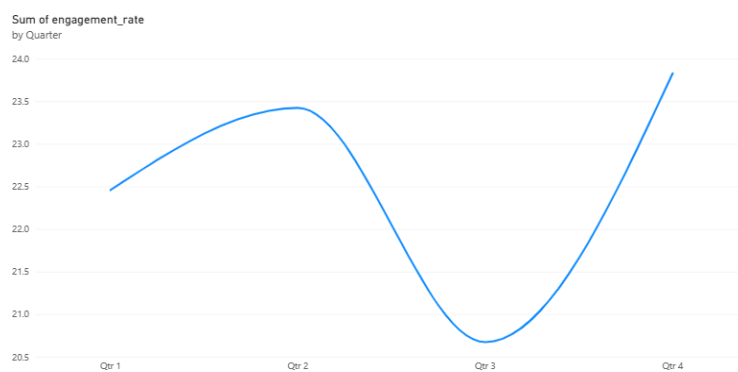


1. Executive Canvas:

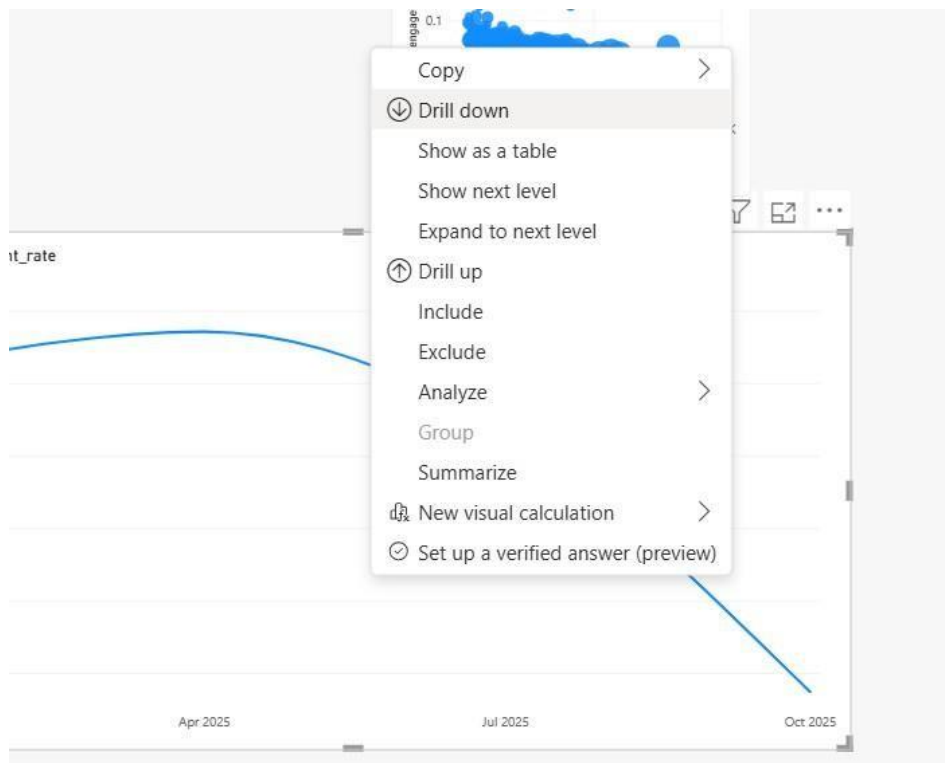
Place Card Visuals across the top: Total Goals, Total Shots, Total Assists, Average Pass Accuracy



Add a Line Chart for sum of goals scored by match week .



Enabled Drill Down

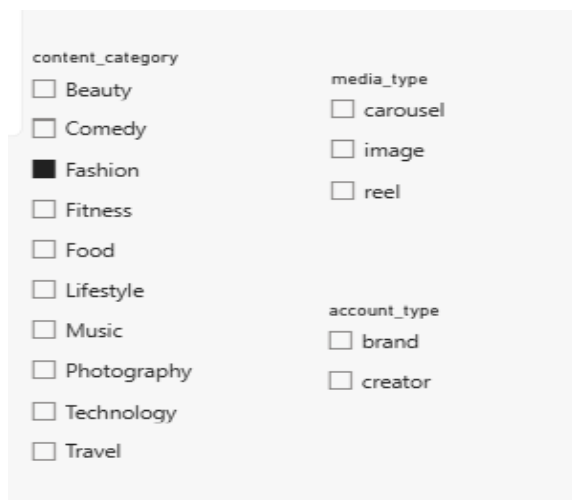


count of matches by team visualised by donut chart

Count of match_id
by team_type

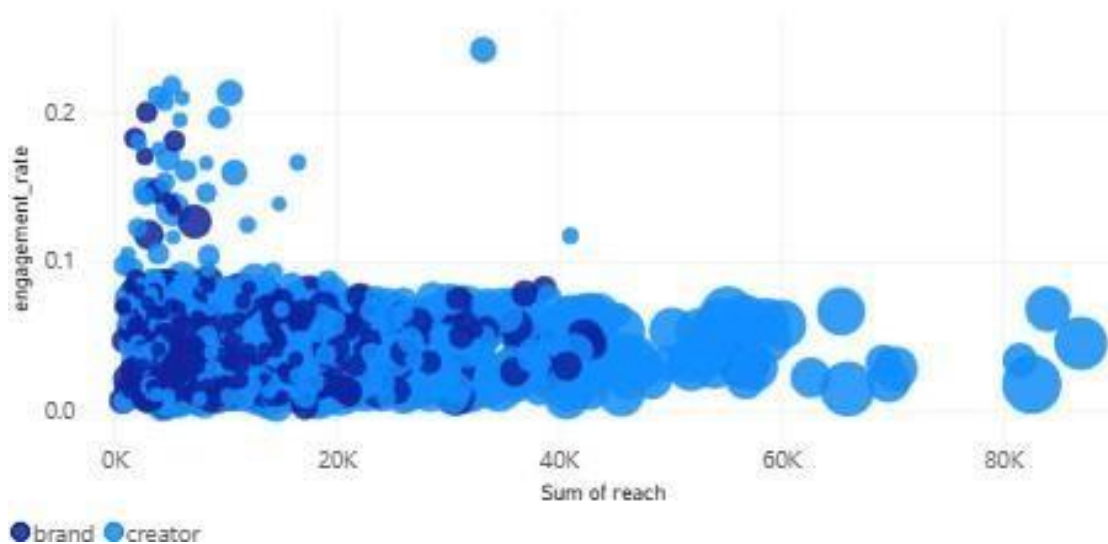


Create 3 slicers based on team, position and competition

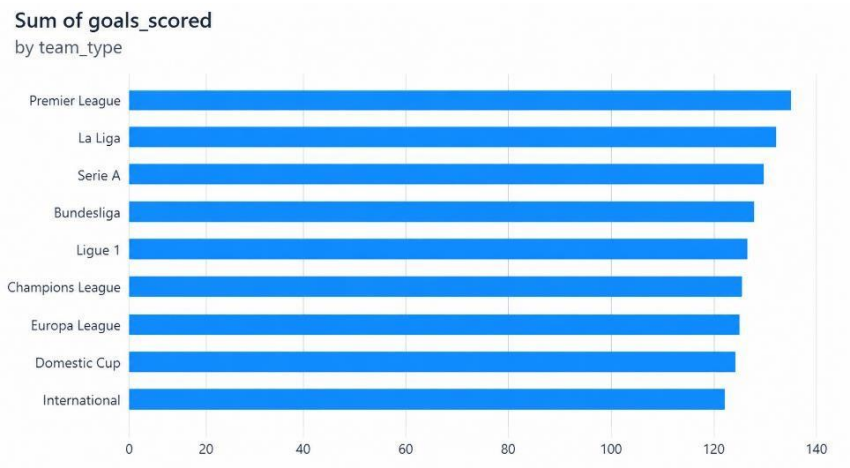


Scatter plot for sum of shots and sum of goals

Sum of reach and Sum of follower_count
by account_type and engagement_rate



average pass accuracy by position



Enable drill-through

Values

Add data fields here

Drill through

Cross-report ☒

Keep all filters ☒

account_type ☐ ☐ ☐

is (All)

match_id	team_type	opponent	match_type	league	goals_scored	shots	shots_on_target	possession	passes	pass_accuracy	xG
M000001	Home Team	Manchester United	League Match	Premier League	2	15	7	58	512	87	1.82
M000002	Away Team	Liverpool	League Match	Premier League	1	12	5	46	398	81	1.05
M000003	Home Team	Chelsea	Cup Match	FA Cup	3	18	9	62	534	89	2.14
M000004	Away Team	Arsenal	League Match	Premier League	0	8	3	41	321	76	0.62
M000005	Home Team	Tottenham	League Match	Premier League	2	14	6	55	476	85	1.48
M000006	Away Team	Manchester City	League Match	Premier League	1	10	4	48	402	82	1.10
M000007	Home Team	Newcastle	League Match	Premier League	4	20	11	63	589	90	2.80
M000008	Away Team	Aston Villa	League Match	Premier League	1	9	4	44	365	79	0.95
M000009	Home Team	Brighton	League Match	Premier League	2	16	8	57	521	86	1.76
M000010	Away Team	West Ham	League Match	Premier League	0	7	2	39	298	74	0.40
M000011	Home Team	Everton	League Match	Premier League	3	17	9	60	548	88	2.21
M000012	Away Team	Leicester City	League Match	Premier League	1	11	5	46	412	81	1.12
M000013	Home Team	Fulham	League Match	Premier League	2	13	6	52	469	84	1.63
M000014	Away Team	Crystal Palace	League Match	Premier League	1	9	4	43	350	78	0.88
M000015	Home Team	Brentford	League Match	Premier League	2	14	7	56	503	86	1.69
M000016	Away Team	Wolves	League Match	Premier League	0	6	2	38	274	71	0.35
M000017	Home Team	Bournemouth	League Match	Premier League	3	19	10	61	560	89	2.45
M000018	Away Team	Southampton	League Match	Premier League	1	8	3	42	330	77	0.71
M000019	Home Team	Nottingham Forest	League Match	Premier League	2	15	7	54	482	85	1.59
M000020	Away Team	Leeds United	League Match	Premier League	1	10	4	45	391	80	0.97
Total					32	254	120	1023	8,276	1,608	25.34

TABLEAU

Import the data from sql server

Tableau - Book1

File Data Server Help

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Open

To a File

- Microsoft Excel
- Text file
- JSON file
- Microsoft Access
- PDF file
- Spatial file
- Statistical file
- More...

To a Server

- MySQL
- Oracle
- Amazon Redshift
- Microsoft SQL Server
- More...

Microsoft SQL Server

General Initial SQL Advanced

Server localhost

Database dashboardDB

Authentication Use Windows Authentication (preferred)

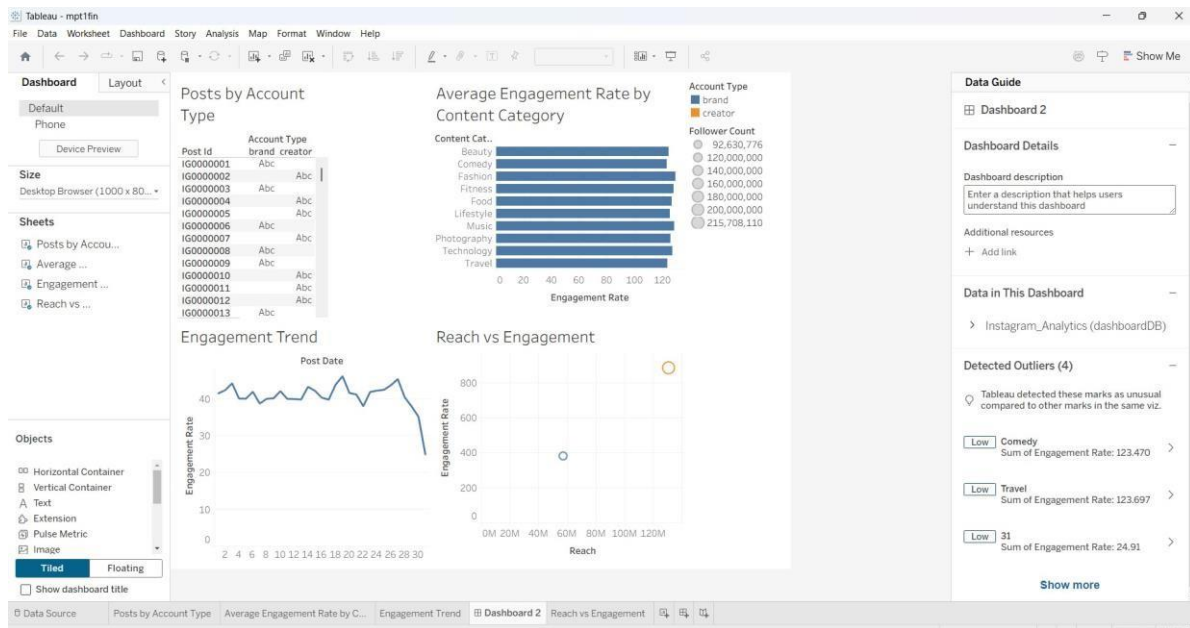
☐ Require SSL

☐ Read uncommitted data

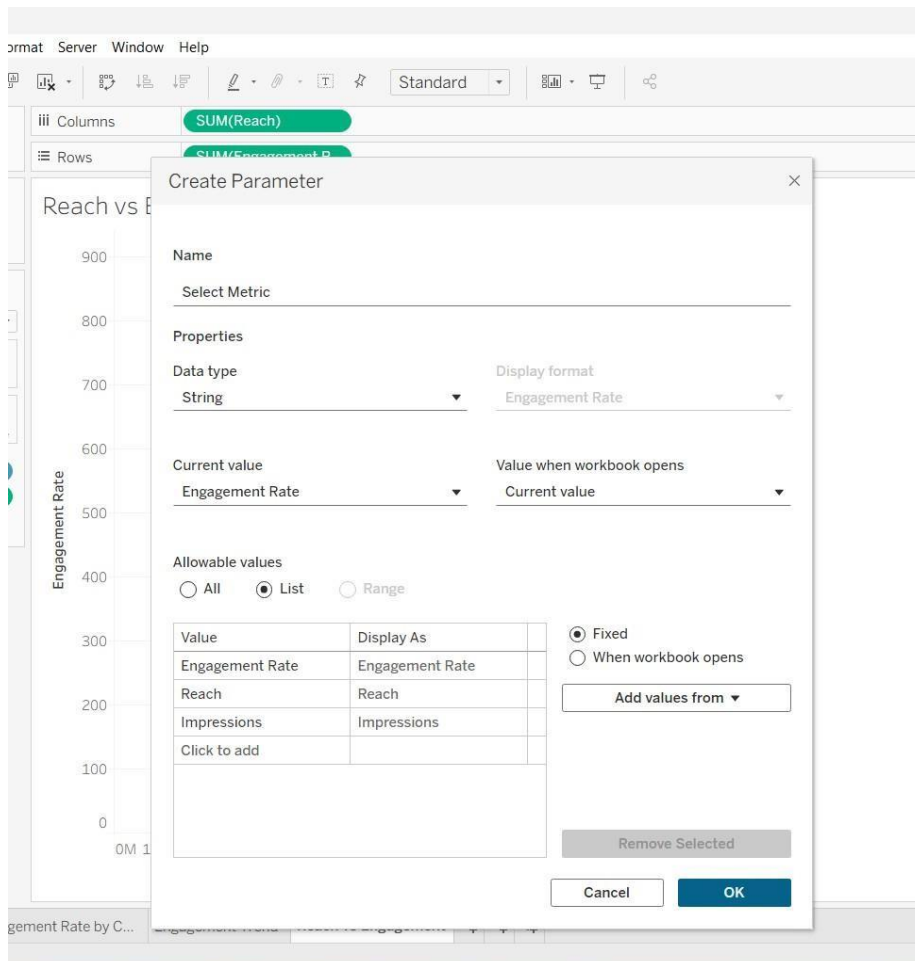
Sign In

Quick Start

FINAL DASHBOARD :



Build parameters



Selected Metric

×

```
CASE [Select Metric]
WHEN "Engagement Rate" THEN [engagement_rate]
WHEN "Reach" THEN [reach]
WHEN "Impressions" THEN [impressions]
END
```

The calculation is valid.

Apply

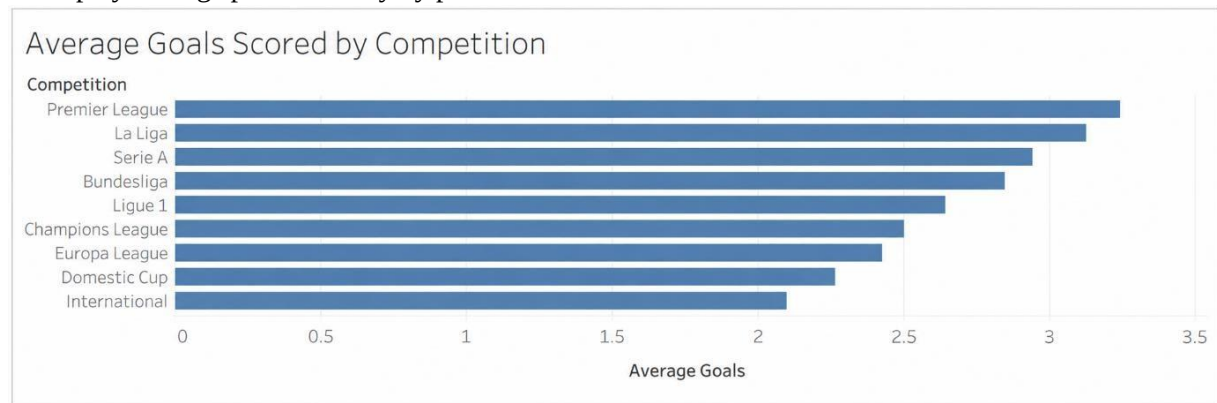
OK

Create a table to display matches by competition

Matches by Team Type

Match Id	Team Type	
	Home Team	Away Team
M0000001	Home Team	Away Team
M0000002	Away Team	Home Team
M0000003	Home Team	Away Team
M0000004	Away Team	Home Team
M0000005	Home Team	Away Team
M0000006	Away Team	Home Team
M0000007	Home Team	Away Team
M0000008	Home Team	Away Team
M0000009	Away Team	Home Team
M0000010	Home Team	Away Team
M0000011	Away Team	Home Team
M0000012	Home Team	Away Team
M0000013	Away Team	Home Team

to display average pass accuracy by position

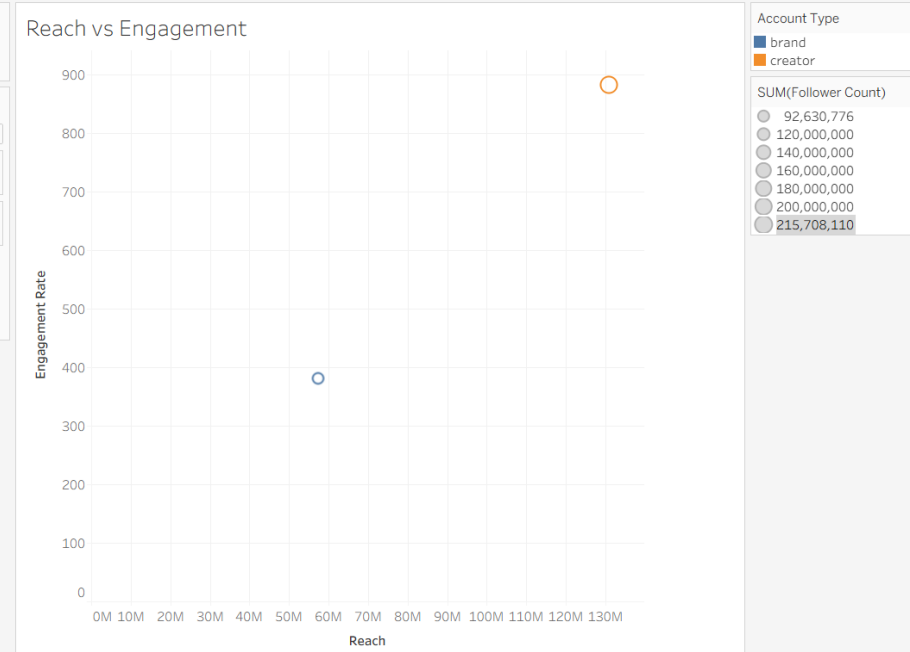


Line chart to visualise goals trend with drill down



Year	2015
Quarter	Q2
Month	May
▪ Day	8
More	▶

Shots VS Goals



Observed Insights

1. The dashboard contains approximately 30K football matches with a total of 82K goals scored. The average pass accuracy is approximately 81%.
This indicates a large volume of match data and strong attacking output across the season.
2. Home teams contribute a larger share of wins than away teams.
Home teams account for approximately 58.4% of wins, while away teams account for about 41.6%.
This shows a clear home-ground advantage in the dataset.
3. Pass accuracy varies across player positions, with Midfielders showing the highest accuracy. Defenders and Goalkeepers also show relatively high accuracy compared with other positions. Forwards appear comparatively lower in the displayed chart, reflecting higher-risk attacking passes.
4. Goal-scoring rate changes across the quarters, reaching its highest level in Q4. The trend increases from Q1 to Q2, decreases during Q3, and rises again in Q4.

This indicates noticeable variation in attacking performance throughout the season.